

Exercise 1 – Replication, Reliability, and Quorums

A team operates a Kafka cluster with $N = 5$ brokers. Each broker has an **independent** probability $p = 0.02$ of being unavailable during any given hour.

Part A. A topic is created with replication factor $r = 3$ and `min.insync.replicas = 2`.

- (a) Compute the probability that all 3 replicas of a partition are unavailable simultaneously (i.e. probability of data *loss* during one hour).
- (b) Compute the probability that the partition becomes *read-only*—that is, fewer than 2 replicas are available (writes blocked) but at least 1 is up (no loss). Provide the numerical value.
- (c) Compute the probability that the partition is *fully writable* (at least 2 of 3 replicas up).

$$(a) \quad p(\text{loss}) = p^r = 0.02^3 = 8 \times 10^{-6}$$

$$\begin{aligned} (b) \quad P(\text{exactly 2 unavailable}) &= \binom{3}{2} \times (0.02)^2 \times (1-0.02)^1 \\ &= \frac{3!}{2!(3-1)!} \times 0.0004 \times 0.98 \\ &= 0.001176 \end{aligned}$$

$$\begin{aligned} (c) \quad P(\text{fully writable}) &= 1 - P(\text{all 3 unavailable}) - P(\text{exactly 2 unavailable}) \\ &= 1 - 0.000008 - 0.001176 \\ &= 0.998816 \end{aligned}$$

Part B. The team also runs a Cassandra cluster with $N = 5$ replicas of every key. They want *strong consistency*, i.e. every read must see the latest write.

- (d) State the inequality that R and W must satisfy.
- (e) List **all** pairs (R, W) with $1 \leq R, W \leq 5$ that satisfy strong consistency.
- (f) Among those pairs, which one minimises the *maximum* of R and W (i.e. best worst-case latency)?

$$(d) \quad R + W > N \rightarrow R + W > 5$$

$$(e) \quad R = 1 \quad (1, 5)$$

$$R = 2 \quad (2, 4), (2, 5)$$

$$R = 3 \quad (3, 3), (3, 4), (3, 5)$$

$$R=4 \quad (4, 2), (4, 3), (4, 4), (4, 5)$$

$$R=5 \quad (5, 1), (5, 2), (5, 3), (5, 4), (5, 5)$$

if) the pair (3,3)

Exercise 2 – Hash Partitioning, Throughput, and Little's Law

An e-commerce company streams clickstream events into Kafka. The producer sends events with a key equal to the `user_id`. There are $P = 8$ partitions in the topic.

Part A. Suppose $n = 800,000$ user events are sent during one hour, with *uniformly distributed* `user_ids`. Model partition assignment as the balls-and-bins model with a perfect hash.

- Compute the expected number of events per partition $\mathbb{E}[X_i]$ and the standard deviation $\sigma(X_i)$.
- Compute the coefficient of variation $\sigma(X_i)/\mathbb{E}[X_i]$ and comment on whether the load is well-balanced.
- Suppose now that 40% of all events come from a single “power user.” What is the load of the partition holding that user’s events (in expectation)? Compare with the average load. What is the practical consequence?

Part B. The consumer group has C consumers and the team measures an end-to-end latency of $L = 80$ ms per event. Each consumer can process **at most** 4,000 events per second.

- The producer rate is $\lambda = 20,000$ events/second. What is the minimum number of consumers C needed to keep up with this rate, given $P = 8$ partitions?
- Using Little’s Law, compute the average number of events “in flight” in the pipeline at steady state.
- The team decides to increase the producer rate to $\lambda' = 40,000$ events/second. They double the number of consumers from 5 to 10, but observe *no improvement* in throughput. Explain mathematically.

$$(a) \quad \mathbb{E}(X_i) = \frac{n}{P} = \frac{800000}{8} = 100000$$

$$V(X_i) = \frac{n}{P} \left(1 - \frac{1}{P}\right) = 100000 \times \left(1 - \frac{1}{8}\right) = 87500$$

$$\sigma(X_i) = \sqrt{V(X_i)} = \sqrt{87500} \approx 295.80$$

$$(b) \quad CV = \frac{\sigma(X_i)}{\mathbb{E}(X_i)} = \frac{295.80}{100000} = 0.002958$$

The coefficient of variation is extremely close to 0.

The relative load imbalance is minimal and the data is very well balanced across the partitions.

(C) Expected load:

$$\frac{800000 \times 0.4 = 320000}{}$$

which will all hash to one specific

partition.

The remaining 480000 events will be distributed uniformly across all 8 partitions (averaging 60000 per partition)

Expected load: $320000 + 60000 = 380000$ events

Average load: 100000 events.

The specific partition's load is 3.8 times higher than the average.

Practical Consequence:

This breaks the uniform assumption and creates a "hot partition". The single partition will bottleneck the system, completely defeating the benefits of hash partitioning.

$$(d) \quad C = \frac{\Lambda}{4000} = \frac{20000}{4000} = 5 \text{ consumers}$$

$$(e) \quad \bar{N} = \Lambda \cdot L = 20000 \times 0.08 = 1600 \text{ events}$$

$$(f) \quad \Lambda_{\max} = \min(P, C) \cdot \Lambda_{\text{per-consumer}}$$

$$= \min(8, 10) \times 4000$$

$$= 8 \times 4000$$

$$= 32000 \text{ events/sec}$$

doesn't improve.

Because Kafka's parallelism is strictly capped by the number of partitions ($P=8$), adding consumers beyond the number of partitions yields no benefit.